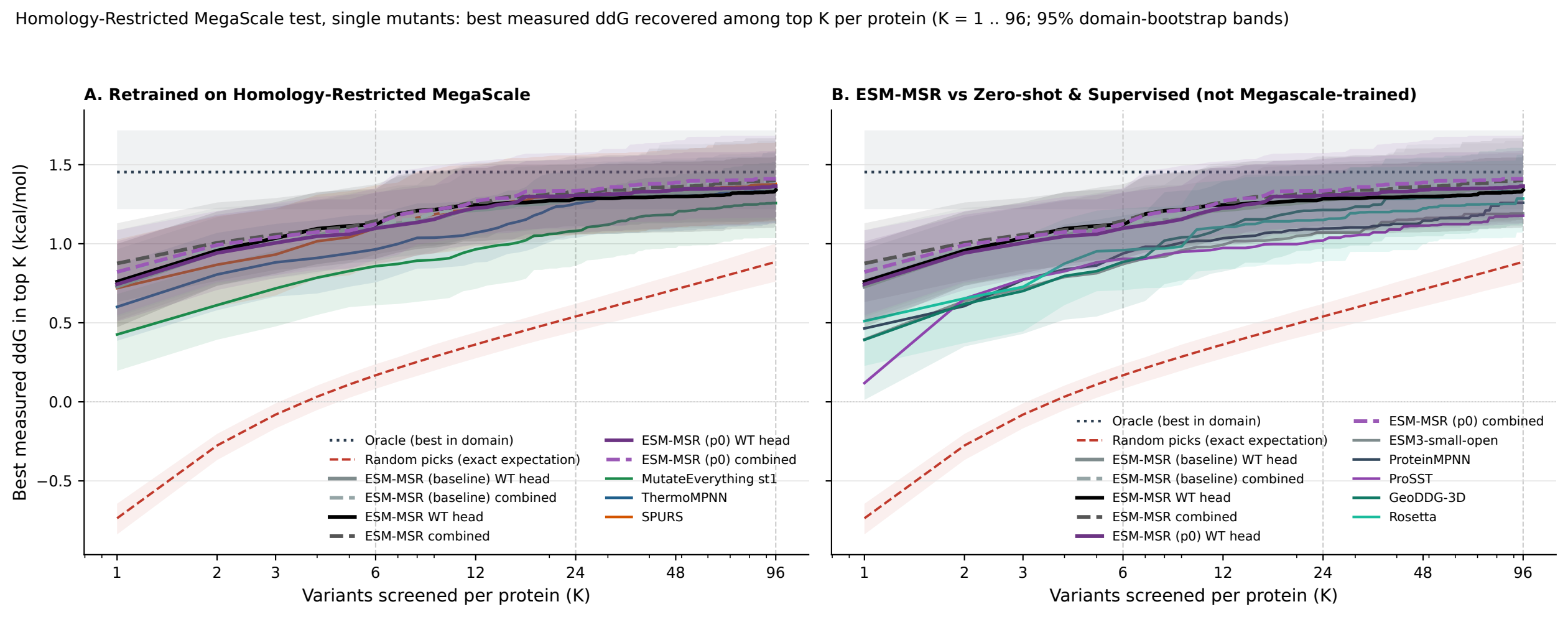

Homology-Restricted MegaScale test, single mutants: best measured ddG recovered among top K per protein (K = 1 .. 96; 95% domain-bootstrap bands)
A. Retrained on Homology-Restricted MegaScale
B. ESM-MSR vs Zero-shot & Supervised (not Megascale-trained)
Best measured ddG in top K (kcal/mol)
1.5
1.0
0.5
0.0
−0.5
1
2
3
6
12
24
48
96
Variants screened per protein (K)
Oracle (best in domain)
Random picks (exact expectation)
ESM-MSR (baseline) WT head
ESM-MSR (baseline) combined
ESM-MSR WT head
ESM-MSR combined
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
MutateEverything st1
ThermoMPNN
SPURS
ESM3-small-open
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta